The Icing on the Cake
The Effects of Monetary Incentives on Income Data Quality in the SIPP

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Abstract

Accurate measurement of key income variables plays a crucial role in economic research and policy decision-making. However, the presence of item nonresponse and measurement error in survey data can cause biased estimates. These biases can subsequently lead to sub-optimal policy decisions and inefficient allocation of resources. While there have been various studies documenting item nonresponse and measurement error in economic data, there have not been many studies investigating interventions that could reduce item nonresponse and measurement error. In our research, we investigate the impact of monetary incentives on reducing item nonresponse and measurement error for labor and investment income in the Survey of Income and Program Participation (SIPP). Our study utilizes a randomized incentive experiment in Waves 1 and 2 of the 2014 SIPP, which allows us to assess the effectiveness of incentives in reducing item nonresponse and measurement error. We find that households receiving incentives had item nonresponse rates that are 1.3 percentage points lower for earnings and 1.5 percentage points lower for Social Security income. Measurement error was 6.31 percentage points lower at the intensive margin for interest income, and 16.48 percentage points lower for dividend income compared to non-incentive recipient households. These findings provide valuable insights for data producers and users and highlight the importance of implementing strategies to improve data quality in economic research.

Keyword*:* SIPP, nonresponse, measurement error, monetary incentives, earnings, asset income, retirement income, experiment

JEL Classification: C82, C83

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1 Introduction

High-quality data obtained through surveys are essential to economic research, determining the accuracy and validity of policy evaluations and theoretical insights. Recent studies (Meyer, Mok, Sullivan, 2015; Kapteyn and Ypma, 2007; Bollinger and Hirsch, 2013; Bollinger et al. 2019) have raised concerns over declining data quality in the survey industry, necessitating more effective strategies to address the issue. The decline in data quality manifests in various ways, including lower response rates (unit nonresponse), refusal to answer specific survey questions (item nonresponse), and the provision of inaccurate information by participating respondents (measurement error).

In an effort to enhance survey data quality and address concerns regarding item nonresponse and measurement errors, the U.S. Census Bureau has adopted several interventions. For instance, the Survey of Income and Program Participation (SIPP) has implemented numerous improvements in the 2014 panel including questionnaire design modifications, integration of survey and administrative data for imputation, and refining data collection processes. Along with these advancements, the Census Bureau conducted a randomized monetary incentive experiment in Waves 1 and 2 to induce unit response (Westra, Sundukchi, Mattingly, 2015). The experiment sought to induce unit response by randomly assigning households to receive either $0, $20, or $40 debit card upon survey completion.[1] The incentives varied between the waves, with households divided into four distinct groups based on the amount received.

In this study, we extend the analysis initiated by Ayromloo and Wilkin (2022) on the impact of monetary incentives on data quality. While our focus is not on the varying amounts of monetary incentives, we examine the overall effect of being assigned to an incentive-receiving group on the

[1] More precisely, the condition for incentive receipt was a "sufficient partial interview," which in SIPP involves the completion of the survey through the assets content and not necessarily beyond that.

quality of data collected[2]. Specifically, we build on their work by not only analyzing the influence of these conditional monetary incentives on item nonresponse rates for earnings but also extending our examination to include critical income types such as interest income, dividend income, rental income, retirement income, Old-Age, Survivors, and Disability Insurance (OASDI) Social Security Administration (SSA) income, as well as benefits from Supplemental Nutrition Assistance Program (SNAP). Furthermore, we introduce a novel dimension to this research by exploring the effects of these incentives on measurement error. These areas are vital for a comprehensive understanding of households' economic well-being and assessing the efficacy of public policies.

Building on our focus to analyze the impact of monetary incentives on various critical income types' nonresponse rates and measurement error, we explore identifying mechanisms through which monetary incentives possibly affect survey outcomes. Insights into these mechanisms empower survey administrators to strategically cultivate desired respondent behaviors, potentially even without monetary incentives. One such mechanism is the consultation of financial records when providing data. By referencing financial records directly, respondents are less likely to omit or misreport information regarding their finances. This direct consultation inherently mitigates the risks of both nonresponse and measurement error for these specific income types. Additionally, the accuracy of personal details, which enhances assignment of unique the Protected Identification Key (PIK) used for record linkage with administrative data, serves as another mechanism. A higher PIK rate suggests respondents provide details more meticulously,

[2] It's worth noting that prior research by Ayromloo and Wilkin (2022) indicated that among the different incentive amounts, only the $40 incentive showed evidence of reducing the earnings nonresponse rate when compared to no incentives. The difference in response rates between the $20 and $40 incentive groups and $20 and no incentive groups was not statistically significant, suggesting that the specific incentive amount might not be an important determinant of response rates. However, the overall presence of an incentive did lower earnings nonresponse, an aspect that this study aims to investigate further and expand to other income types using the Correlated Random Effects (CRE) model.

which in turn reduces chances of item nonresponse for various income questions and minimizes discrepancies between survey and administrative data.

Our findings shed light on how these mechanisms operate in practice. Conditional monetary incentives have a positive effect on data reliability. We observe that for incomes such as interest, dividend, and rental, there is an increase in financial record consultation as indicated by the interviewers during the data collection process. Further enhancing data quality, there is improved provision of accurate personal details. The probability of respondents consulting their financial records increases by 1.5 percentage points when they are assigned to an incentive-receiving group. Being part of an incentive-receiving household also increases the probability of an individual being assigned a unique PIK for linkage to administrative data, which is highly connected to whether household members provide accurate information on their full name and exact date of birth.

We observe that the nonresponse rate for earnings data is lowered by 1.3 percentage points, while the nonresponse rate for OASDI income falls by 1.51 percentage points when respondents are assigned to incentive-receiving groups. For income types such as interest income, dividend income, rental income, and retirement income, the results remain inconclusive, and no discernable influence of incentive assignment on nonresponse rates can be deduced. The same holds true for SNAP coverage nonresponse rates and earnings measurement error.

Notably, and in line with prior discussion regarding the consultation of financial records, there is a marked reduction in measurement error for interest and dividend incomes when respondents belong to incentive-receiving groups. Specifically, interest income measurement error is reduced by 6.31 percentage points at the intensive margin, while dividend income measurement error falls by 16.48 percentage points. These results reaffirm the notion that direct consultation of

financial records, prompted by monetary incentives, can effectively drive the reduction in measurement error for these specific income types. Conversely, there is no clear evidence suggesting that incentive assignment reduces measurement error in rental income, retirement income, OASDI income, or SNAP benefits.

This paper contributes to the literature on measurement error and survey methodology by focusing on the effects of incentives on data quality for critical economic variables, using administrative data as a benchmark. Our findings suggest that incentives can have a positive impact on data quality for certain variables, with additional evidence that the incentives increased respondent effort and motivation. By shedding light on the benefits and limitations of using incentives in surveys, our research can inform the design of more effective survey methodologies to improve the quality of data collected and, consequently, the reliability of economic research.

2 Literature Review

Our work is connected to both the literature documenting the extent of measurement error for income variables and the literature on incentives. For income measurement error, the primary focus has been on earnings. Many papers focus on matching the Current Population Survey Annual Social and Economic Supplement (CPS ASEC) to earnings data from the Social Security Administration (SSA), such as Lillard et al. (1986), Bound and Krueger (1991), Bollinger and Hirsch (2013), and Bollinger et al. (2019). These papers document extensive differences for a large fraction of the sample, with larger differences for individuals with very low and very high amounts of earnings. Large differences in earnings are found in other surveys as well, such as the SIPP (Chenevert, Klee, and Wilkin, 2016). There have been studies on measurement error for other

income sources, such as asset income (Eggleston and Reeder 2018), social security and retirement income (Bee and Michell 2017), and SNAP receipt (Celhay et al. 2022), also documenting sizable amounts of measurement error in these sources. For the most part, many of these papers document the extent of measurement error without proposing solutions to the problem. Eggleston and Reeder (2018) provide evidence that encouraging respondents to consult financial records during the interview improves data quality, and Celhay et al. (2022) examine error in SNAP receipt across multiple surveys to shed some light on how survey design choice may affect error.

Our research dovetails with the extensive survey methods literature focusing on incentives. Many studies predominantly investigate the impact on response rates, including notable works by Church (1993), James and Bolstein (1992), Mercer et al. (2015), and Hsu et al. (2017). However, a subset of this literature delves deeper, exploring how incentives influence the quality of responses. For instance, Davern et al. (2003) in their analysis of the 1996 SIPP panel conclude that incentives did not affect data quality, as measured by item nonresponse rates and the prevalence of entirely missing data for household members. This observation of null effects is echoed by studies like Shaw et al. (2001).

Conversely, some research identifies positive effects of incentives. Goetz et al. (1984) observe reduced item nonresponse and more extensive answers to open-ended questions. Cole et al. (2015) find a reduction in straight-lining behaviors in a survey of college students. On the other hand, a few studies pinpoint negative effects. Barge et al. (2012) determine that students who

received a $15 incentive exhibited increased item nonresponse rates, shorter interview durations, and more straight-lining behavior. A unique perspective is offered by Stecklov et al. (2018), who analyze data from developing countries and find that respondents in the incentives condition report much lower consumption and income. The authors surmise this behavior might stem from a strategy to seem more financially strained, possibly in hopes of attracting future survey opportunities or assistance.

In reviewing the literature, it is evident that most studies concerning data quality and incentives typically use indirect methods to gauge data quality. Common metrics include straight-lining behaviors and the duration spent on each question. Additionally, these studies often rely on qualitative measurements, frequently utilizing Likert scales, to determine variations like the extent of straight lining between different incentive scenarios. Notably, there is a discernable gap in this body of research: the consideration of responses to financial queries involving specific dollar amounts. Such questions could entail distinct response behaviors and introduce unique measurement error dynamics.

Our paper makes several notable contributions. Firstly, we specifically focus on income generated through labor (earnings), assets, retirement, and SNAP (a social safety net program). These categories represent highly sensitive and private questions that are crucial for comprehending economic well-being. Both item nonresponse and measurement error are relatively

high for these variables, so our setting is one in which interventions to improve data quality can be especially impactful.

Secondly, we thoroughly examine various aspects of data quality, encompassing item nonresponse and measurement error. To evaluate measurement error, we use administrative data equivalents that directly pertain to the variables of interest. This approach provides a more precise and direct method for assessing survey data quality, rather than relying on less direct methods such as incidence of straight-lining and interview times.

Thirdly, we establish a clear causal link between being assigned to an incentive receiving group and the rates of item nonresponse and measurement error within a longitudinal setting. Our analysis takes into account observed and unobserved individual characteristics that could potentially bias the results. By controlling for these factors, we ensure the robustness and reliability of our findings.

Lastly, our paper presents distinct and compelling evidence on a mechanism that connects incentives to data quality — respondent effort. As discussed above, prior research has primarily analyzed outcomes such as straight-lining and time spent per question, which could serve as proxies for both survey quality and respondent effort, leading to potential conflations and imprecise inferences. By contrast, our data and methodology advance the literature by offering a more refined and targeted way of evaluating survey data quality and separating that from understanding respondent effort through their look-up of financial records in answering assets and

retirement income questions and providing more accurate personally identifiable information for PIK assignment and linkage to the administrative data.

3 Theory

The effects of monetary incentives on data quality is ex-ante ambiguous. Potential reasons exist for both detrimental and beneficial impacts on the quality of survey responses. Detrimental effects may arise when respondents, motivated by the incentive, rush through the survey, leading to less accurate answers. Incentives for in-person surveys might also induce socially desirable responses, as respondents may seek to please the interviewer or avoid appearing uncooperative.

Conversely, incentives might improve response accuracy by fostering feelings of reciprocity and appreciation, resulting in more cooperative and engaged respondents. Additionally, the fear of forfeiting the incentive due to a lack of seriousness might push respondents to provide more accurate answers.

Leverage-Saliency Theory (Groves, Singer, and Corning 2000) offers a useful framework to examine these competing effects. According to this theory, individuals weigh perceived benefits against the costs of participating in a survey. This evaluation applies to both the decision to participate and the decision to provide accurate answers, conditional on participation.

We can categorize individuals into two groups based on their reaction to incentives. The first group includes individuals who are driven by the monetary incentive to participate (unit response). However, their reluctance to participate might still manifest in higher item nonresponse or lower quality responses. The second group comprises individuals already inclined to participate without incentives but lack motivation to provide accurate responses. The incentive can increase their sense of reciprocity, leading to more accurate responses.

The overall observed effect of randomized monetary incentives on data quality depends on whether the sampled households predominantly belong to the first or the second group.

4 Data

We use the 2014 Panel of the Survey of Income and Program Participation (SIPP), which follows a nationally representative sample of households over four years, conducting annual interviews. This panel's primary objectives, consistent with previous SIPP panels, were to track short-term dynamics in employment, income, household composition, and government assistance program participation and eligibility. The household sample, drawn from the universe of non-institutionalized U.S. population, included anoversampling from high-poverty areas to yield more precise program-related estimates.

Notably, the 2014 SIPP panel introduced several changes compared to prior panels. It reduced the frequency of interviews from three times to once annually and incorporated a calendar tool in the survey instrument, allowing respondents to align events like employment and marital status changes with specific dates. Additionally, there were some alterations in the survey's content (please see Eggleston and Gideon (2017) for an evaluation of changes to the 2014 panel's wealth content).

A key survey attribute in the 2014 SIPP was the implementation of a randomized monetary incentive experiment during the first two waves, aimed at improving unit response rates and reducing nonresponse bias. Historically, the SIPP has used monetary incentives in various forms during different waves of 1996, 2001, 2004, 2008, and 2014 panels to enhance unit response rates (please refer to Westra, Sundukchi, and Mattingly (2015) for an analysis of these outcomes in 2014 and details of incentive distributions in previous panels).

For the 2014 panel, households were randomly assigned to one of four incentives groups before the survey's commencement, as outlined in Table 1. The allocation of incentives, disbursed

through debit cards along with separate PIN numbers (Westra, Sundukchi, and Mattingly (2015)) was contingent upon either completing the survey or providing a "sufficient partial" interview in Waves 1 and 2 (see footnote (1) for the definition of "sufficient partial" interview in the SIPP context). Group 1 households were slated to receive no incentives in both waves. Group 2 households were allocated $0 in Wave 1 and $40 in Wave 2. Those in Group 3 were set to receive $20 in Wave 1 and $0 in Wave 2. Group 4 households initially received $40 in Wave 1, then split into subgroups 4a and 4b for Wave 2, receiving $40 and $0 respectively.

To evaluate the effect of these conditional monetary incentives on item nonresponse rates and measurement error for various income types, we link the SIPP data at the individual level using numerous administrative data. SIPP respondents are linked to administrative data via the U.S. Census Bureau's Person Identification Validation System (PVS), as described in Wagner and Layne (2014). This procedure matches both survey data and administrative data to a master reference file. Individuals who are matched are given a unique identifier called a Protected Identification Key (PIK), which acts as an anonymized social security number that can be used to link administrative datasets and surveys. For 2014 SIPP Wave 1, 91.9 percent of individuals over 15 years old were assigned a PIK (Eggleston and Reeder 2018). The topics we pick are ones where we have both survey and administrative data measures for. Because most of our administrative data is at the annual level, we sum the SIPP variable to the annual level, even though some of the variables are available at the month level.

Our earnings administrative data source is the Detailed Earnings Record (DER) of the Master Earnings File from the Social Security Administration. The DER is a longitudinal history of annual earnings going back to 1978, which contains both administrative employee and self-employed earnings. For employees, the DER contains data from every W-2 a person received. We

match at the individual level, so we sum over all DER earnings to create our administrative data variable. For the self-employed with positive earnings, the DER contains person-level earnings from reported Form 1040, Schedule SEs (1040-SE). In our measurement error analyses, we combine employer earnings and self-employed earnings together.

For OASDI (Old-Age, Survivors, and Disability Insurance) social security income, we also use SSA data, this time taking data from the Master Beneficiary Record (MBR). These data contain information on receipt of retirement, survivors, or disability benefits, containing various data relevant to the construction of monthly benefit amounts. Data on retirement benefits from pensions and 401(k) withdrawals, for example, comes from Internal Revenue Service (IRS) Form 1099-R. SNAP data comes from state agencies. For the references years in question (2013 and 2014), the U.S. Census Bureau has data from only 20 states, so the SNAP analyses are restricted to those states[3].

The Individual Master File of all IRS 1040 tax returns contains data on interest, dividend, and gross rental income. Interest income corresponds to taxable interest and tax-exempt interest from lines 8a and 8b of the 1040 form. Dividend income consists of ordinary and qualified dividends from lines 9a and 9b. Gross rental income consists of rental and royalty income from lines 3 and 4 of the Schedule E form. For most observations, the IRS variable is taken from one matched 1040 record. For a small number of married couples, each spouse is matched to a different 1040 record. We have concluded that in most of these cases, the married partners are filing separate tax returns. Because of this, we add up the values from the two different returns when creating the combined income totals for a married couple.

[3] Those states are AZ, CT, FL, HI, ID, IN, KY, MD, MI, MN, MS, MT, NV, NJ, ND, OR, TN, UT, and WY.

In terms of conceptual alignment, the survey and administrative data mostly measure the same concepts. The main difference is with retirement income. The 2014 SIPP did not have explicit questions on IRA and 401(k) withdrawals, which are captured in our 1099-R data. Many people may have reported these withdrawals as pension income. But nevertheless, it is likely many people with 401(k) withdrawals are not reporting them on SIPP. Additionally, the administrative data do not capture low amounts of income in certain circumstances. People with asset income under $10 may not be reporting this on tax returns given banks are not required to send 1099-INTs in these cases. For earnings, unincorporated self-employed workers are only required to file a 1040-SE if their profits exceed $400, so individuals with low or negative profits will not show up in the DER.

Table 2 presents a detailed overview of the summary statistics for our key variables, divided into three panels. Panel A deals with variables that shed light on how monetary incentives influence item nonresponse and measurement error for various income variables. Panel B focuses on item nonresponse variables and Panel C is dedicated to measurement error variables.

Panel A reveals that about 25 percent of survey participants consulted financial records to answer questions about their assets and wealth, which includes the income, dividend, and rental income types discussed in this paper. Furthermore, Panel A shows that 93 percent of all survey participants had high enough quality in their personally identifiable information, such as name, and date of birth, that they were able to be assigned a PIK.[4]

Panel B shows that in our sample of 53,500 survey participants who were asked about earnings, 22 percent either declined to respond or did not know the answer, resulting in imputed responses. Moreover, among various income categories, interest income and dividend income have

[4] Additional, respondents can only be assigned a PIK if they have a social security number or an Individual Taxpayer Identification Number, which excludes some proportion of immigrants.

the highest nonresponse rates at about 50 percent and 59 percent, respectively. Additionally, the number of participants asked about rental income is the lowest among all variables, even lower than those asked about having received SNAP benefits in the reference year. The nonresponse rates for rental income, retirement income, and OASDI are 19 percent, 17 percent, and 17 percent, respectively. The nonresponse rate for SNAP benefits is 0.9 percent, with 4,300 survey participants who were asked about SNAP benefits.

An important caveat to highlight is that certain item nonrespondents answer range follow-up inquiries, where they are asked whether their income falls into certain brackets. This mechanism offers constraints on potential imputed values, narrowing the gap between imputed survey data and its corresponding administrative records.

Finally, Panel C sets the stage by providing baseline estimates of measurement errors. This lays the foundation to gauge the magnitude of shifts in measurement errors from being randomly assigned to an incentive-receiving household. Panel C documents sizable measurement error for many of our variables. For earnings, we find a median error value of $5,165, meaning the typical survey respondent reports an earnings value that is over $5,000 different than what we find in administrative records. Error in retirement income is also high with a median of $5,232. The SNAP type 2 error of 28.7% means that almost 30% of respondents who receive SNAP do not report this receipt in SIPP. In summary, there is an economically significant amount of measurement error in SIPP that incentives could possibly affect.

5 Empirical Framework

To estimate the average treatment effect (ATE) of incentive assignment on data quality, we use a regression framework that accounts for potential sources of endogeneity. We begin with the following baseline regression equation:

$$Y_{it} = \alpha + \beta I_{it} + \varepsilon_{it}, \qquad (1)$$

where Y_{it} represents the outcome variable of interest, such as income nonresponse or income measurement error for various types of income for individual i at year t. When Y_{it} represents nonresponse, it equals to 1 if the income type is imputed and 0 otherwise[5]. When Y_{it} represents a measurement error, then it's defined at the intensive margin by the inverse hyperbolic sine transformation of the absolute value of the difference between the survey's (SIPP) and corresponding administrative records' values, where both are positive, *i.e.,* $sinh^{-1}|(SIPP\ amounts - administartive\ amounts)|$, where $SIPP\ amounts > 0$ and $administartive\ amounts > 0$. When Y_{it} represents Type I error in SNAP coverage, it equals to 1 if survey data indicate coverage and administrative data do not, and 0 otherwise. Lastly, when Y_{it} represents Type 2 error in SNAP coverage, it equals to 1 if survey data indicate no coverage and administrative data show coverage, and 0 otherwise. I_{it} is an indicator for incentive assignment, and $\varepsilon_{it}$is a stochastic error term.

Equation (1) assumes strict exogeneity, $E(\varepsilon_{it}|I_{it}) = 0$. While the incentives were randomly assigned, this assumption could be violated due to self-selection into unit response, non-random attrition, or a combination of omitted variable bias and imbalance in the treatment and control conditions. If these confounders are observable, we can directly control for them in the following equation:

$$Y_{it} = \alpha + \beta I_{it} + X'_{it}\gamma + \varepsilon_{it}, \qquad (2)$$

where X_{it} is a vector of time-varying, observable individual characteristics that are correlated with incentive assignment and our outcome variable.

[5] For the sake of completeness, it is important to note that various survey questions may be combined to create the different income types studied in this paper. In this context, $Y_{it} = 1$ if *any* of the components that contribute to its construction is missing.

To account for unobservable individual heterogeneity, we use a model that controls for individual heterogeneity following the preferred specification from Ayromloo and Wilkin (2022):

$$Y_{it} = \alpha_i + \beta I_{it} + X'_{it}\gamma + u_{it}, \qquad (3)$$

where α_i represents individual fixed effects, capturing time-invariant unobservable characteristics. Estimation of this model requires eliminating the individual fixed effects. Two prevalent methods for achieving this are the fixed effects model and the random effects model.

In the fixed effects model, the identifying variation stems from individuals who had a change of assignment between Waves 1 and 2. This means they were assigned to a non-incentive receiving group in Wave 1 and an incentive-receiving group in Wave 2, or vice versa. This method estimates the ATE among switchers, not the overall ATE.

The random effects model assumes that, conditional on observable controls, the incentive assignment is uncorrelated with the error term and unobserved individual heterogeneity. However, this assumption is what we aim to test.

Following the approach of Ayromloo and Wilkin (2022), we use the Mundlak (1978) method, which allows individual fixed effects to depend on the time averages of time-varying covariates:

$$\alpha_i = \psi + \overline{Z'_\iota}\zeta + \eta_i, \qquad (4)$$

where ψ is a time-invariant component common to all individuals, $\overline{Z}_\iota$ is a vector of time averages of time-varying covariates (e.g., I_{it}, X_{it}, and the wave dummy), and η_i is an individual-specific component uncorrelated with the time-varying covariates. Substituting (4) into (3) yields:

$$Y_{it} = \psi + \overline{Z}'_\iota\zeta + \beta I_{it} + X'_{it}\gamma + e_{it}, \qquad (5)$$

where $e_{it} = \eta_i + u_{it}$.

This specification allows us to estimate the effect of incentive assignment on data quality, while controlling for both observable and unobservable individual heterogeneity. With the Mundlak approach, similar to the fixed effects model, the variation used to identify β comes from within-individual changes — in this case, changes in an individual's treatment assignment from being in a non-incentive group to an incentive group, or vice versa. This variation allows us to causally interpret how a change in treatment assignment affects an individual's outcome while adjusting for time-invariant individual characteristics. Additionally, by incorporating time averages of the time-varying covariates, the Mundlak approach provides us with a framework to test for and control for the correlations between these covariates and unobserved individual heterogeneity.

In all specifications, we focus on two primary outcome variables: item nonresponse rates and measurement error at the intensive margin for earnings, investment income from assets, retirement account income, and OASDI income. The nonresponse rate for SNAP pertains to coverage, while measurement error for SNAP addresses benefit amounts. Due to the skewed distribution of these income measures and the presence of zero values in survey and administrative income variables, we construct measurement error outcome variables at the intensive margin where a respondent has positive survey and administrative records and apply an inverse hyperbolic sine transformation to the absolute differences between survey and administrative amounts.

Two recent papers, Chen and Roth (2023) and Mullahy and Norton (2022), address the challenges in estimating ATEs with zero-valued outcomes and discuss the limitations of transformation methods, such as logarithmic and inverse hyperbolic sine transformations, when the outcome variable has a mass at zero. Mullahy and Norton (2022) argue that estimated average marginal effects in transformed models with zeros are biased and may result in large changes in coefficients' point estimates. Chen and Roth (2023) demonstrate that ATEs for the transformed

outcome will be arbitrarily sensitive to units of the untransformed outcome variable and suggest alternative approaches for dealing with zero-valued outcomes.

In our data, we do not have a large mass at zero. However, we continue to examine the sensitivity of our results to the transformation of the intensive-margin measurement error outcomes by using the arc percentage change or relative difference transformation in Section 7 as a robustness check. The arc percentage change transformation involves dividing the absolute value of the simple difference between survey (SIPP) and administrative amounts by the combined survey and administrative amounts' average:

$$\left|2 * \frac{(SIPP\ amounts - administrative\ amounts)}{(SIPP\ amounts + administrative\ amounts)}\right|, \qquad (6)$$

where both SIPP and administrative amounts are positive. Despite the absence of a large mass at zero in our data, this approach ensures that our findings remain reliable even in the presence of potential issues with the inverse hyperbolic sine transformation model.

6 Results

In this section, we delve into our key findings, shedding light on how incentives can influence survey response rates and accuracy. Our analysis unfolds in three subsections, each providing a unique perspective on the incentive effect.

The first subsection details how monetary incentives affect two specific mechanisms for their potential influence on various income nonresponse rates and measurement errors. These mechanisms are the likelihood of respondents consulting their financial records during the interview and the provision of more accurate personally identifiable information for PIK assignment. Next, the second subsection discusses the findings from our item nonresponse analysis. Finally, the third subsection presents our findings on measurement errors.

6.1 Mechanism Results

Before diving into the empirical findings, we succinctly revisit the key mechanisms by which monetary incentives operate. Monetary incentives can motivate respondents to directly consult their financial records. This action inherently reduces the chances of omissions or misreports, thus minizine nonresponse rates and measurement errors for various income questions. Incentives may also lead to a more meticulous provision of personal details, enhancing PIK rates used for record linkage. A heightened PIK rate reduces item nonresponse and discrepancies between survey and administrative data.

With these mechanisms in mind, we delve into our specific findings.

Table 3 presents an analysis exploring how incentives could impact nonresponse rates and measurement error. Specifically, the analysis focuses on whether individuals in incentive-receiving households are more likely to consult financial records when responding to asset income questions.

This table is structured into six columns, each representing a different specification. All specifications include a wave dummy to control for average behavioral differences between the initial and the second wave of the survey. Controls for observable time-varying individual characteristics (X_{it}) are introduced in Columns (3) and (4) — including age, age squared, race indicators, foreign born status indicator, marital status indicators, educational attainment indicators, proxy response indicator, interview mode indicator, and log of household size including only members aged 15 and older. These characteristics are consistently controlled for across all regression tables. Columns (5) and (6) expand the controls to include time averages of time-varying individual characteristics ($\overline{Z}_t$), using the Mundlak device. This approach allows us to control for unobservable individual characteristics that may correlate with the treatment.

The results in Table 3 show that being part of an incentive-receiving household consistently increases the probability of consulting financial records, ranging from 1.24 percentage points increase in the most basic specification (Column 1) to a 1.50-percentage point increase in our preferred specification (Column 6). With an average record use rate of 25.14%, this represents a modest effect size. This preferred specification controls for both observable and unobservable individual characteristics, while applying longitudinal weights. Wald statistics in Columns (5) and (6) indicate that unobservable individual heterogeneity is present but not correlated with the treatment, thus rejecting the null hypothesis.

It is worth noting that our specifications in this table and all the subsequent regression tables alternate between applying cross-sectional and longitudinal weights to the estimates. The latter approach restricts our sample to individuals who participate in the survey across both waves, thus providing a cleaner comparison than cross-sectional weights for the Mundlak specification. The cross-sectional weights include both respondents who stay in the sample for both waves and those who only participated in the initial wave and attritted afterward. However, utilizing longitudinal weights reduces our sample size.

The results in Table 3 suggest that incentives may reduce nonresponse rates and measurement error unless other mechanisms offset the record consultation effect. As item nonresponse rates and measurement errors decrease for many earnings and investment income variables, the consultation of financial records appears to be an important factor driving these results.

Building on the implication derived from Table 3, we delve deeper into understanding the mechanisms by which incentives potentially reduce nonresponse rates and measurement error. In addition to the utilization of financial records, encouraging the provision of accurate personal

details may also improve data quality. For linking respondents to administrative records in SIPP, it is crucial for respondents to provide accurate details on their full name and date of birth. If they do not do this, then respondents are less likely to be assigned a PIK by the Census Bureau's PVS. Therefore, whether a respondent is assigned a PIK provides some indirect information of the likelihood they gave complete name and date of birth information.

In Table 4, we present results on how incentive assignment is connected to the likelihood of PIK assignment. Similar to Table 3, this table features six columns, each representing a different model specification. Columns (1) and (2) control solely for wave effects. Columns (3) and (4) further incorporate controls for time-varying, observable individual characteristics, denoted as X_{it}. Finally, Columns (5) and (6) adhere to the Mundlak specification and control for time averages of the said time-varying covariates, I_{it}, and the wave dummy, denoted by $\bar{Z}_i$. Across these various specifications, the results demonstrate a remarkable consistency. The estimated effect of residing in a household assigned to receive an incentive leads to an increase in the probability of PIK assignment. This increase falls within a narrow range of 0.35 to 0.44 percentage points. To put these numbers into context, consider that the base rate of PIK non-assignment is 7.04 percent. Thus, the effect of 0.35 to 0.44 percentage points implies a reduction in the likelihood of PIK non-assignment by approximately 5.0 – 6.3 percent, which is a meaningful effect size. Just as we have seen with financial record consultation, the assignment of PIKs emerges as another important area through which incentives contribute to data quality enhancements.

6.2 Nonresponse Results

Next, we turn our attention to nonresponse rates, evaluating the effects of incentives on earnings, five other types of income (interest, dividends, rental, retirement, and SSA OASDI income), and SNAP receipt.

Table 5 offers an in-depth look at the impact of living in an incentive-receiving household on earnings nonresponse rates. The table uses eight different specifications, with each one considering the earnings nonresponse rate as the dependent variable.

Specifications vary across the table's columns. Columns (1) and (2) are simple models, only controlling for a wave dummy. Columns (3) to (8), however, add controls for time-varying individual characteristics (X_{it}). These include the same variables as Table 3 with additional controls for receiving hourly wages, extra earnings, and business earnings, along with the log of job count, and log of administrative earnings in the DER. Columns (5) to (8) include controls for time averages of time-varying individual characteristics ($\overline{Z}_{\iota}$), while Columns (7) and (8) condition on DER earnings being positive.

Across all specifications, assignment to an incentive-receiving group consistently lowers the earnings nonresponse rate. This effect ranges from 1.2 percentage points in Column (7) to 1.47 percentage points in Column (6). The estimated effects remain remarkably consistent, falling within a tight range.

The Wald statistics in specifications (5) to (8) test the hypothesis that unobserved individual characteristics are correlated with the treatment. In all these columns, the Wald statistics reject this null hypothesis, indicating that unobserved individual heterogeneity is present in our data but not correlated with our treatment. Thus, these results provide robust evidence of the impact of incentives on the earnings nonresponse rate.

Table 6 outlines our analysis results on the effects of assignment to an incentive-receiving group on various other types of income nonresponse rates. The table is divided into five panels. Each panel focuses on a different income type: interest income (Panel A), dividend income (Panel B), rental income (Panel C), retirement income (Panel D), and SSA OASDI income (Panel E).

Each panel has six columns. These columns condition on the corresponding survey investment income being positive, and they incorporate a wave dummy and controls for time-varying individual characteristics (X_{it}) and time averages of time-varying individual characteristics ($\bar{Z}_i$). Columns (3) to (6) include an additional control for the inverse hyperbolic sine transformation ($sinh^{-1}$) of corresponding administrative income. Moreover, Columns (5) and (6) apply a condition where both SIPP and corresponding administrative income data are positive. This allows us to control for potential skewness at the extremes of investment income levels and mitigates sensitivity to transformation due to large number of zero values.

Across all columns, we utilize our preferred Mundlak specification to control for both observable and unobservable individual heterogeneity potentially correlated with the treatment. Moreover, the Wald statistic in each column tests the hypothesis that unobserved individual characteristics correlate with the treatment.

Panel A shows that residing in an incentive-receiving household consistently reduces the interest income nonresponse rate between 2.30 and 2.50 percentage points, as seen in Columns (1) to (4). However, when we condition for both SIPP and administrative interest income being positive in Columns (5) and (6), our sample size substantially shrinks. This restriction is particularly relevant for interest income, where banks only have to send tax forms to customers if they have $10 or more of interest income, which may lead to underreporting on individual 1040 tax returns for individuals with small amounts of interest income. This reduction in sample size limits our statistical power to the point where we cannot reject the null hypothesis that incentive assignment does not decrease the interest income nonresponse rate. Nevertheless, the magnitude and direction of the coefficients remain roughly consistent with prior estimates. The Wald statistics

confirm the presence of unobserved heterogeneity in our sample but indicate no correlation with our treatment effect.

Panel B presents a similar pattern, with the difference being a smaller number of observations for dividend income. Residing in an incentive-receiving household reduces the dividend income nonresponse rate between 2.36 and 2.79 percentage points in Columns (1) to (4). Yet, with only 5,000 and 4,400 observations in Columns (5) and (6), we lack the statistical power to reject the null hypothesis that incentives do not affect the dividend income nonresponse rate. In all columns, the Wald statistics show the existence of unobserved individual heterogeneity in our sample, but it's not correlated with our treatment, except in Columns (2), (4) and (6) with longitudinal weights, where we cannot reject the null hypothesis that unobserved individual heterogeneity is correlated with the treatment.

In Panels C and D, we find no evidence to reject the null hypothesis that assignment to an incentive-receiving group affects the nonresponse rates for rental or retirement income. The coefficients are small relative to their standard errors but are comparable and in the same direction as those in previous panels.

In contrast, Panel E shows statistically significant effects for SSA income. Living in an incentive-receiving household consistently reduces the SSA income nonresponse rate between 1.44 and 1.58 percentage points. As with the previous panels, the Wald statistics confirm the existence of unobserved heterogeneity in our sample and also demonstrate no correlation with the treatment effect.

Table 7 details our analysis results on the effects of residing in an incentive-receiving household on the nonresponse rate for SNAP coverage. The table is organized into four columns, each providing different perspectives based on our preferred Mundlak specification.

Columns (1) and (2) are our basic conditions of someone receiving SNAP in SIPP, which always includes people with positive benefits amount in the survey. We further refine this condition in Columns (3) and (4), requiring both survey and administrative SNAP benefits amounts to be positive. These latter columns also introduce a control for administrative SNAP amounts.

Despite our various specifications, the coefficients are small relative to our standard errors. We lack the statistical power to reject the null hypothesis stating that assignment to an incentive-receiving group does not affect the SNAP coverage nonresponse rate. We also fail to reject any correlation between unobserved individual heterogeneity and our treatment effect. Given the very low rates of item nonresponse for SNAP receipt, this result is not too surprising.

6.3 Measurement Error Results

In the preceding subsection, we examined the role of incentives in affecting nonresponse rates. Our attention now pivots to examine the accuracy of the elicited responses. Similar to the results presented in the previous subsection, all results in this subsection alternate across columns between using cross-section weights and longitudinal weights, unless stated otherwise.

Table 8 presents the results of our analysis on the effect of residing in an incentive-receiving household on earnings measurement error at the intensive margin. For clarity, it's important to recall that, as previously described in this paper, earnings measurement error is defined as the inverse hyperbolic sine transformation ($sinh^{-1}$) of the absolute value of the difference between SIPP earnings and the earnings amount reported in the DER, and because we are focusing on the intensive margin, we condition on both SIPP and DER earnings being positive.

The table is divided into six columns, each corresponding to a unique specification. Every specification includes a wave dummy and controls for observable individual characteristics (X_{it}).

Columns (3) to (6) extend the controls by including the inverse hyperbolic sine transformation of the DER earnings amounts. This consideration is based on the hypothesis that earnings measurement error might vary across different income levels. Subsequently, Columns (5) and (6) further control for unobserved individual heterogeneity by including controls for time averages of time-varying covariates ($\overline{Z}_\iota$).

The coefficients across all specifications are concentrated within a narrow range, between -0.74 and -1.01 percentage points. However, these coefficients, in relation to their standard errors, are quite small, which leads us to not reject the null hypothesis that incentives do not affect earnings measurement error. The Wald statistics for Columns (5) and (6) reject the null hypothesis that unobserved individual heterogeneity is correlated with our treatment.

Following a structure analogous to Table 6, Table 9 delves into the effects of residing in an incentive-receiving household on the measurement error of various types of income at the intensive margin, rather than focusing on nonresponse rates. Each of the five panels in the table reflects the intensive-margin measurement error for a different type of income: interest income (Panel A), dividend income (Panel B), rental income (Panel C), retirement income (Panel D), and SSA OASDI income (Panel E).

Each panel is further segmented into six columns, each corresponding to a unique specification. All specifications incorporate a wave dummy, and those in Columns (3) to (6) also include controls for observable individual characteristics (X_{it}). Controls for time averages of time-varying individual characteristics ($\overline{Z}_\iota$) are added to Columns (5) and (6). All specifications take into account the inverse hyperbolic sine transformation ($sinh^{-1}$) of the corresponding administrative income. This allows for our coefficient to be roughly interpreted as a percent change.

Panel A consistently demonstrates across all specifications that living in a household assigned to an incentive-receiving group leads to a reduction in the measurement error of interest income at the intensive margin, by between 6.20 and 7.52 percentage points. In terms of absolute dollar amounts, this translates to a reduction in measurement error of approximately $7.87 to $9.55, based on a median base rate of $127 in measurement error as shown in Table 2. While the effect in percentage points may seem large, this dollar-value framing underscores the modest absolute impact of the treatment. The Wald statistics test the null hypothesis that unobserved individual heterogeneity is correlated with the treatment, and in this instance, fails to reject it.

A similar trend is observable in Panel B, which reveals that across all specifications, living in an incentive-receiving household lowers dividend income measurement error at the intensive margin by between 13.26 and 16.48 percentage points. Here too, Wald statistics in Columns (5) and (6), fail to reject correlation between any unobserved individual heterogeneity with our treatment.

Panels C, D, and E depict a more mixed picture. In these panels, the results are too noisy to reject the null hypothesis that residing in an incentive-receiving household does not influence rental income measurement error, retirement income measurement error, or SSA OASDI income measurement error, all at the intensive margin, respectively.

Table 10 delves into the effects of residing in an incentive-receiving household on SNAP benefit amounts' measurement error at the intensive margin, as well as on two types of SNAP coverage reporting errors. All columns in this table present our preferred Mundlak specification, allowing for the control of both observable and unobservable individual heterogeneity. Moreover, each column incorporates a wave dummy to account for the two waves of the experiment.

In the first two columns, we report on SNAP benefits' measurement error at the intensive margin, where we condition on having positive SNAP benefits per both SIPP and state-level administrative records. In these columns, we also condition on the inverse hyperbolic sine transformation of administrative SNAP benefit amounts. This measure helps us control for potential correlations between misreporting of coverage and the actual amount of SNAP benefits received as per administrative records.

Columns (3) and (4) tackle Type I SNAP coverage error — where respondents falsely report receiving SNAP benefits on SIPP, despite having no record of benefits in administrative data. For these analyses, we condition our results on the absence of SNAP coverage in administrative records.

Columns (5) and (6) handle Type II SNAP coverage error — where respondents falsely report not receiving SNAP benefits on SIPP, while administrative data confirm their receipt of these benefits. In these instances, our analysis conditions on the presence of SNAP coverage in administrative data.

Across the board, our findings are noisy, and we can't reject the null hypothesis that assignment to an incentive-receiving group does not affect SNAP benefits measurement error or either type of reporting error.

Wald statistics in Columns (1) to (4) confirm the presence of unobserved heterogeneity in our sample but also reject any correlation between this unobserved heterogeneity and our treatment. However, we are unable to reject the hypothesis that unobserved heterogeneity is corelated with our treatment effect in Columns (5) and (6).

7 Robustness Results

In this section, we address recent concerns raised by Chen and Roth (2023) and Mullahy and Norton (2022) regarding the use of inverse hyperbolic sine transformed outcome variables when the outcome variable frequently appears at zero. As a robustness check, we re-estimate our measurement error regressions for earnings, asset income types, and SNAP benefits using the arc percentage change or relative difference formulation as defined in Equation (6). This formulation caters to situations where the dependent variables include zero values and skewed distributions. It handles zero in either survey or administrative data (or both), without leading to undefined or infinite values, and is less sensitive to extreme values, thus reducing the skewness impact.

Table 11 presents the impact of residing in a household assigned to receive an incentive on the relative earnings error to the average earnings (arc percentage change earnings error) where an individual has both survey and administrative records across eight distinct specifications. This section's goal is to ascertain our results' robustness, and thus, we report results ranging from our most basic to our preferred specification, similar to the intensive-margin earnings measurement error results in Table 8. However, Table 11 includes two additional columns, where Columns (1) and (2) exclude controls for time-varying observable individual characteristics. All specifications incorporate a wave dummy and condition on SIPP earnings being nonnegative. Columns (3) and (4) still exclude any time-varying observable individual characteristics but condition on the inverse hyperbolic since transformation of administrative earnings in the DER. This condition is maintained in all subsequent specifications.

Even though Columns (1) – (4) lack equivalents in Table 8, they provide insight into how each additional control or condition may affect the estimated coefficients, and that no specification produces statistically significant results for earnings measurement error. Columns (5) and (6)

introduce controls for time-varying observable individual characteristics and align with Columns (3) and (4) in Table 8, respectively. Columns (7) and (8), the Mundlak specifications, add controls for time averages of time-varying covariates and correspond to Columns (5) and (6) in Table 8. Consistent with previous regression tables in this paper, the estimates in odd columns use cross-sectional weights, while those in even-numbered columns use longitudinal weights.

Consistently across all specifications, we observe that point estimates are small relative to their standard errors, and we cannot reject the null hypothesis that incentives do not impact earnings measurement error. As before, the Wald statistics confirm the presence of unobserved individual heterogeneity in our data but reject any correlation between this unobserved individual heterogeneity and our treatment.

Table 12 parallels Table 9 in the presentation of the effects of residing in an incentive-receiving household on the intensive-margin measurement error for various income types: interest income (Panel A), dividend income (Panel B), rental income (Panel C), retirement income (Panel D), and SSA OASDI income (Panel E). The distinguishing element in this table compared to Table 9 is the construction of the dependent variable. Here, we use the arc percentage change of each income type's error as the dependent variable.

Following the pattern established in Table 11, Table 12 also gradually introduces more complex specifications to further examine the robustness of our results. The first two columns include only a wave dummy and do not have direct equivalents in Table 9. However, Columns (3) and (4) in this table are equivalent to Columns (1) and (2) in Table 9. Here, we incorporate a wave dummy and control for the inverse hyperbolic sine transformation of the corresponding administrative income type amount. Columns (5) and (6) introduce controls for time-varying observable individual characteristics, aligning with Columns (3) and (4) in Table 9. Lastly,

Columns (7) and (8) present results from the Mundlak specifications, similar to Columns (5) and (6) in Table 9, including time averages of time-varying covariates.

Estimates in Panel A are in the same direction as Table 9. The estimates fall in a tight range of -2.25 to -2.34. This is about a third of the estimates we find in Table 8, although the magnitude is not directly comparable given the different ranges of the dependent variable. In our preferred Mundlak specifications, Columns (7) and (8), we observe that residing in a household assigned to receive monetary incentive lowers measurement error at the intensive margin for interest income between 2.34 to 2.30 percentage points. In Panels B - E, the estimates' small size in comparison to their standard errors precludes us from rejecting the null hypotheses that assignment to an incentive-receiving group does not affect dividend income, rental income, retirement income, and SSA OASDI income, respectively. The only variable whose significance is changed is dividend income, which has significant coefficients in Table 9 but insignificant results in these robustness results for Table 12.

Table 13 presents the impact of living in a household assigned to receive incentives on the arc percentage change error for SNAP benefit amounts at the intensive margin. For comparability purposes with Table 10, all models in this table follow Mundlak specifications, accounting for wave effects, observable individual characteristics, and time averages of time-varying covariates.

The first two columns do not control for the inverse hyperbolic sine transformation of administrative SNAP benefit amounts. In contrast, Columns (3) and (4) incorporate this control, making them equivalent to Columns (1) and (2) in Table 10. Similar to our findings in Table 10, there is no discernable evidence suggesting that incentives influence the measurement error for SNAP benefits amounts. Nonetheless, the Wald statistics continue to affirm the presence of

unobserved individual heterogeneity within our sample, while also dismissing any correlation between this unobserved heterogeneity and our treatment effect.

8 Conclusion

In conclusion, our paper examines the effect of randomized monetary incentives on reducing nonresponse rates and measurement errors in survey data. While the primary objective of these incentives was to enhance unit response rates and diminish nonresponse bias, our findings reveal an "extra layer": the impact these incentives have on reducing nonresponse rates and measurement errors in certain income types.

We explore mechanisms possibly driving this effect, such as an increased tendency to consult financial records and improved provision of personal information for administrative linkage. Our results show that incentives can lower nonresponse rates and measurement error for some types of income. For nonresponse rates, we find significant effects for earnings and SSA OASDI Income. For measurement error, we find significant effects for interest and dividend income. Yet, these effects are not uniformly observed across all income types. For some forms of income, such as rental and retirement income, we found the data to be noisy, and no statistically significant effect was discernable. We also turn our attention to SNAP coverage and benefits amounts, where we were unable to reject the null hypotheses that incentives do not affect SNAP coverage nonresponse rates, benefits measurement error, or the prevalence of two types of reporting errors.

Our findings highlight the complex, multifaceted relationship between incentives, respondent characteristics, and the nature of various income sources. While monetary incentives have shown potential in reducing nonresponse rates and measurement error for some income types, our analysis reveals that they do not uniformly achieve these reductions across all income types.

Therefore, it is evident that a one-size-fits-all approach to administering conditional monetary incentives may not be optimal for the goal of universally reducing nonresponse rates and measurement error.

Our study contributes to the ongoing discourse on enhancing data quality in income surveys. The insights provided here should inform future survey designs and methodologies. While assessing the overall worth of survey incentives extends beyond the scope of this paper, we have expanded upon the item nonresponse dimension initially introduced by Ayromloo and Wilkin (2022), and have introduced a new dimension for consideration: the impact of these incentives on measurement error. These factors, alongside unit response rates, nonresponse bias, and interview costs, among others, should be integral to evaluating the overall benefits of survey incentives.

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Tables

Table 1. Monetary Incentives Across Groups

Group	Wave 1	Wave 2	Observations	Percent
1	$0	$0	14,000	24.1
2	$0	$40	14,500	25.0
3	$20	$0	14,500	25.0
4a	$40	$40	12,000	20.7
4b	$40	$0	3,100	5.3

Source: Authors' calculations from the Survey of Income and Program Participation 2014, Waves 1 and 2.

Notes: Observations refer to unique persons aged 15 and older whose records are not entirely imputed for a given wave and incentive group.

Table 2. Summary Statistics

Variables	Mean	Standard Deviation	Median	Observations
A. Mechanisms				
Financial records look-up	0.2514	0.4338		87,000
PIK assignment	0.9296	0.2559		96,500
B. Nonresponse				
Earnings	0.2177	0.4127		53,500
Interest income	0.4997	0.5000		32,500
Dividend income	0.5903	0.4918		7,200
Rental income	0.1862	0.3893		3,100
Retirement income	0.1668	0.3728		10,500
OASDI SSA income	0.1681	0.3740		22,000
SNAP benefits	0.0095	0.0971		4,300
C. Measurement error	Inverse Hyperbolic Sine		Levels	
Earnings	9.152	1.728	$5,165	49,500
Interest income	5.673	2.591	$127	13,000
Dividend income	7.561	2.621	$1,317	5,000
Rental income	7.535	3.595	$2,496	2,200
Retirement income	8.661	2.46	$5,232	9,500
OASDI SSA income	8.411	1.247	$2,686	22,000
SNAP benefits	6.259	1.791	$333	2,600
	Percentage			
SNAP Type I error	0.0140	0.1177		26,000
SNAP Type II error	0.2870	0.4524		5,400

Source: Authors' calculations from the Survey of Income and Program Participation 2014, Waves 1 and 2, the Detailed Earnings Records, the 1040 tax returns, the 1099-R, the Master Beneficiary Record, and states' Supplemental Nutrition Assistance Program records in 2013 and 2014. Survey weights were used to construct these statistics.

Notes: For all variables, the sample is restricted to individuals aged 15 and older whose records are not entirely imputed, that is, they are not Type Z individuals. Panels B and C are conditional on the respondent having a PIK, or for the asset variables, either the respondent or the spouse having a PIK. Panel B is also conditional on the either the SIPP income variable being greater than zero (for income variables) or receiving SNAP in SIPP (for SNAP benefits). Panel C for the income and SNAP benefits amounts is conditional on both the SIPP and administrative data variable being greater than zero. For SNAP Type I error, the sample is restricted to no SNAP benefits in administrative data and for Type II error, the sample is restricted to SNAP benefits in administrative data. SNAP variables are also restricted to individuals who reside in a state with SNAP administrative data coverage.

Table 3. Asset Record Use Regression

	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
Incentive	0.0124***	0.0129***	0.0144***	0.0149***	0.0144***	0.0150***
	(0.0041)	(0.0044)	(0.004)	(0.0043)	(0.004)	(0.0043)
Controls						
Wave 2	Yes	Yes	Yes	Yes	Yes	Yes
X_{it}			Yes	Yes	Yes	Yes
$\bar{Z}_{\iota}$					Yes	Yes
Weights						
Cross sectional	Yes		Yes		Yes	
Longitudinal		Yes		Yes		Yes
Observations	87,000	73,000	87,000	73,000	87,000	73,000
Clusters	28,500	21,500	28,500	21,500	28,500	21,500
Wald statistic					5.22***	4.85***

Source: Authors' calculations from the Survey of Income and Program Participation 2014, Waves 1 and 2.
Notes: Sample is restricted to individuals aged 15 and older with non-missing values to the asset record use question. Sample also restricted to respondents whose records are not entirely imputed, that is, they are not Type Z individuals. The specifications that control for time-varying observable individual characteristics include wave dummy, proxy response indicator, interview mode indicator, log number of household members aged 15 and older, age, age squared, educational attainment indicators, and marital status indicators. Standard errors are in parentheses and clustered at the household level. Sample size and number of clusters are rounded according to Census disclosure avoidance policies. ****p≤0.01; **p≤0.05; *p≤0.1*.

Table 4. PIK Assignment Regression

	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
Incentive	0.0040*	0.0035*	0.0044**	0.0041**	0.0044**	0.0041**
	(0.0021)	(0.0021)	(0.0020)	(0.0020)	(0.0020)	(0.0020)
Controls						
Wave 2	Yes	Yes	Yes	Yes	Yes	Yes
X_{it}			Yes	Yes	Yes	Yes
$\bar{Z}_\iota$					Yes	Yes
Weights						
Cross sectional	Yes		Yes		Yes	
Longitudinal		Yes		Yes		Yes
Observations	96,500	80,500	96,500	80,500	96,500	80,500
Clusters	29,500	22,500	29,500	22,500	29,500	22,500
Wald statistic					6.16***	14.03***

Source: Authors' calculations from the Survey of Income and Program Participation 2014, Waves 1 and 2, and the Census Bureau's Person Identification Validation System (PVS) reference file.

Notes: Sample is restricted to individuals aged 15 and older. Sample is also restricted to individuals whose records are not entirely imputed, that is, they are not Type Z individuals. The specifications that control for time-varying observable individual characteristics include wave dummy, proxy response indicator, interview mode indicator, log number of household members aged 15 and older, age, age squared, educational attainment indicators, and marital status indicators. Standard errors are in parentheses and clustered at the household level. Sample size and number of clusters are rounded according to Census disclosure avoidance policies. ****p≤0.01; **p≤0.05; *p≤0.1*.

Table 5. Earnings Nonresponse Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Variables								
Incentive	-0.0121***	-0.0129***	-0.0131***	-0.0146***	-0.0133***	-0.0147***	-0,0120***	-0.0130***
	(0.0045)	(0.0047)	(0.0044)	(0.0046)	(0.0044)	(0.0046)	(0.0045)	(0.0047)
Controls								
Wave 2	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
X_{it}			Yes	Yes	Yes	Yes	Yes	Yes
$\bar{Z}_{\iota}$					Yes	Yes	Yes	Yes
If DER earnings > 0							Yes	Yes
Weights								
Cross sectional	Yes		Yes		Yes		Yes	
Longitudinal		Yes		Yes		Yes		Yes
Observations	53,500	44,500	53,500	44,500	53,500	44,500	50,500	42,000
Clusters	21,000	16,000	21,000	16,000	21,000	16,000	20,500	15100
Wald statistic					6.40***	3.74***	5.64***	3.28***

Source: Authors' calculations from the linked Survey of Income and Program Participation 2014, Waves 1 and 2, with the Detailed Earnings Records in 2013 and 2014.

Notes: Sample is restricted to individuals aged 15 and older. It excludes individuals with entirely imputed records (Type Z), imputed jobs, and who do not have a PIK. But we include respondents who worked for an employer, were self-employed, or had another work arrangement. In addition to the core set of time-varying observable individual characteristics that are common across every regression table (wave dummy, proxy response indicator, interview mode indicator, log number of household members aged 15 and older, age, age squared, educational attainment indicators, and marital status indicators), all earnings regressions including the ones reported in this table include log number of jobs, and indicators for hourly pay, extra earnings, and business earnings. Specifications in this table also include the log of earnings in the DER. Standard errors are in parentheses and clustered at the household level. Sample size and number of clusters are rounded according to Census disclosure avoidance policies. ****p≤0.01; **p≤0.05; *p≤0.1*

Table 6. Asset and Retirement Income Nonresponse Regressions

	(1)	(2)	(3)	(4)	(5)	(6)
A. Interest income						
Incentive	-0.0246***	-0.0230***	-0.0250***	-0.0234***	-0.0130	-0.0066
	(0.0065)	(0.0069)	(0.0065)	(0.0069)	(0.0097)	(0.0103)
If SIPP interest income > 0	Yes	Yes	Yes	Yes	Yes	Yes
$sinh^{-1}$ admin interest income			Yes	Yes	Yes	Yes
If SIPP & admin interest income amounts > 0					Yes	Yes
Observations	32,500	27,000	32,500	27,000	13,000	11,000
Clusters	19,000	15,000	19,000	15,000	8,400	6,600
Wald statistic	3.11***	1.54*	3.68***	2.34***	2.90***	2.06***
B. Dividend income						
Incentive	-0.0237*	-0.0279**	-0.0236*	-0.0278**	-0.0101	-0.0204
	(0.0127)	(0.0137)	(0.0127)	(0.0137)	(0.0150)	(0.0159)
If SIPP dividend income > 0	Yes	Yes	Yes	Yes	Yes	Yes
$sinh^{-1}$ admin dividend income			Yes	Yes	Yes	Yes
If SIPP & admin dividend income amounts > 0					Yes	Yes
Observations	7,200	6,200	7,200	6,200	5,000	4,400
Clusters	5,100	4,100	5,100	4,100	3,500	2,800
Wald statistic	2.12***	1.08	1.99**	1.02	1.53*	0.95
C. Rental income						
Incentive	-0.0129	-0.0134	-0.0121	-0.0125	-0.0070	-0.0095
	(0.0158)	(0.0165)	(0.0157)	(0.0165)	(0.0173)	(0.0182)
If SIPP rental income > 0	Yes	Yes	Yes	Yes	Yes	Yes
$sinh^{-1}$ admin rental income			Yes	Yes	Yes	Yes
If SIPP & admin rental income amounts > 0					Yes	Yes
Observations	3,100	2,600	3,100	2,600	2,200	1,900
Clusters	2,100	1,700	2,100	1,700	1,500	1,200
Wald statistic	1.23	0.64	1.12	0.67	2.10***	1.89**

D. Retirement income						
Incentive	-0.0137	-0.0116	-0.0131	-0.0110	-0.0107	-0.0101
	(0.0087)	(0.0088)	(0.0087)	(0.0088)	(0.0092)	(0.0092)
If SIPP retirement income > 0	Yes	Yes	Yes	Yes	Yes	Yes
$sinh^{-1}$ admin retirement income			Yes	Yes	Yes	Yes
If SIPP & admin retirement income amounts > 0					Yes	Yes
Observations	10,500	9,400	10,500	9,400	9,500	8,400
Clusters	6,000	4,900	6,000	4,900	5,200	4,300
Wald statistic	3.98***	1.86**	3.79***	1.77**	4.27***	2.18***
E. SSA income						
Incentive	-0.0147**	-0.0146**	-0.0146**	-0.0144**	-0.0158**	-0.0151**
	(0.0064)	(0.0066)	(0.0064)	(0.0066)	(0.0064)	(0.0066)
If SIPP SSA income > 0	Yes	Yes	Yes	Yes	Yes	Yes
$sinh^{-1}$ admin SSA income			Yes	Yes	Yes	Yes
If SIPP & admin SSA income amounts > 0					Yes	Yes
Observations	22,000	19,500	22,000	19,500	22,000	19,500
Clusters	10,000	8,100	10,000	8,100	10,000	8,100
Wald statistic	6.04***	1.96**	6.10***	2.18***	5.65***	1.93**
Controls						
Include (Wave 2, X_{it}, Z_i) > 0	Yes	Yes	Yes	Yes	Yes	Yes
Weights						
Cross sectional	Yes		Yes		Yes	
Longitudinal		Yes		Yes		Yes

Source: Author's calculations from the linked Survey of Income and Program Participation 2014, Waves 1 and 2, with 1040 tax returns for interest, dividend, a
Record for Old-Age, Survivors, and Disability Social Security Income in 2013 and 2014.

Notes: Sample is restricted to individuals aged 15 and older and includes imputed responses. We also drop individuals with entirely imputed records (Type Z). The
spouse having a PIK, and the other variables are conditional on the respondent having a PIK. All specifications in Panels A – E condition on their corresponding s
individual characteristics (wave dummy, proxy response indicator, interview mode indicator, log number of household members aged 15 and older, age, age squa

each panel also include the inverse hyperbolic sine transformation of corresponding administrative income type amount. Standard errors are in parentheses and cl
Census disclosure avoidance policies. ****p≤0.01; **p≤0.05; *p≤0.1*.

Table 7. SNAP Nonresponse Regression

	(1)	(2)	(3)	(4)
Variables				
Incentive	-0.0040	-0.0060	0.0024	-0.0021
	(0.0044)	(0.0043)	(0.0031)	(0.0013)
Controls				
Wave 2	Yes	Yes	Yes	Yes
Include $(X_{it}, \overline{Z}_t)$	Yes	Yes	Yes	Yes
if SNAP SIPP amount > 0	Yes	Yes	Yes	Yes
$Sinh^{-1}$ SNAP admin amount			Yes	Yes
if SNAP admin amounts > 0			Yes	Yes
Weights				
Cross sectional	Yes		Yes	
Longitudinal		Yes		Yes
Observations	4,300	3,700	3,400	3,000
Clusters	2,000	1,700	1,700	1,400
Wald statistic	0.72	0.74	0.39	0.22

Source: Author's calculations from the linked Survey of Income and Program Participation 2014, Waves 1 and 2, and state-level records on Supplemental Nutrition Assistance Program in 2013 and 2014.

Notes: Sample is restricted to individuals aged 15 and older and includes imputed responses. Sample is also restricted to individuals who reside in a state with SNAP administrative data coverage. We also drop individuals with entirely imputed records (Type Z) and who do not have a PIK. The universe for the first two columns is only the SIPP and is then shrunk to only the linked records between the SIPP and the state-level SNAP records in the last two columns. The time-varying observable individual characteristics in these specifications are consistent with the core set of time-varying covariates used in all the other regression tables (wave dummy, proxy response indicator, interview mode indicator, log number of household members aged 15 and older, age, age squared, educational attainment indicators, and marital status indicators), but the specifications in Columns (5) and (6) also include the inverse hyperbolic sine transformation of state-level administrative SNAP benefit amounts. Standard errors are in parentheses and clustered at the household level. Sample size and number of clusters are rounded according to Census disclosure avoidance policies. ***$p \leq 0.01$; **$p \leq 0.05$; *$p \leq 0.1$.

Table 8. Earnings Measurement Error Regression

	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
Incentive	-0.0086	-0.0074	-0.0101	-0.0100	-0.0099	-0.0098
	(0.0165)	(0.0177)	(0.0159)	(0.0171)	(0.0159)	(0.0170)
Controls						
Include (Wave 2, X_{it})	Yes	Yes	Yes	Yes	Yes	Yes
If SIPP earnings ≥ 0	Yes	Yes	Yes	Yes	Yes	Yes
$\bar{Z}_{\iota}$					Yes	Yes
$sinh^{-1}$(DER earnings amounts)			Yes	Yes	Yes	Yes
Weights						
Cross sectional	Yes		Yes		Yes	
Longitudinal		Yes		Yes		Yes
Observations	49,500	41,500	49,500	41,500	49,500	41,500
Clusters	20,000	15,000	20,000	15,000	20,000	15,000
Wald statistic					8.36***	7.96***

Source: Authors' calculations from the linked Survey of Income and Program Participation 2014, Waves 1 and 2, with the Detailed Earnings Records in 2013 and 2014.

Notes: Sample is restricted to individuals aged 15 and older. It excludes individuals with entirely imputed records (Type Z), imputed jobs, and who do not have a PIK. But we include respondents who worked for an employer, were self-employed, or had another work arrangement. All specifications condition on having positive earnings in both SIPP and the DER. In addition to the core set of time-varying observable individual characteristics that are common across every regression table (wave dummy, proxy response indicator, interview mode indicator, log number of household members aged 15 and older, age, age squared, educational attainment indicators, and marital status indicators), all earnings regressions including the ones reported in this table include log number of jobs, and indicators for hourly pay, extra earnings, and business earnings. Some specifications in this table also include the inverse hyperbolic sine transformation of earnings in the DER. Standard errors are in parentheses and clustered at the household level. Sample size and number of clusters are rounded according to Census disclosure avoidance policies. ****p≤0.01; **p≤0.05; *p≤0.1.*

Table 9. Asset and Retirement Income Measurement Error Regression

	(1)	(2)	(3)	(4)	(5)	(6)
A. Interest income						
Incentive	-0.0752**	-0.0631*	-0.0709**	-0.0620*	-0.0714**	-0.0631*
	(0.0348)	(0.0373)	(0.0345)	(0.0371)	(0.0345)	(0.0370)
$sinh^{-1}$admin interest income	Yes	Yes	Yes	Yes	Yes	Yes
Observations	13,000	11,000	13,000	11,000	13,000	11,000
Clusters	8,400	6,600	8,400	6,600	8,400	6,600
Wald statistic					1.20	1.25
B. Dividend income						
Incentive	-0.1337**	-0.1621**	-0.1326**	-0.1648**	-0.1344**	-0.1648**
	(0.0661)	(0.0708)	(0.0651)	(0.0696)	(0.0651)	(0.0695)
$sinh^{-1}$admin dividend income	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5,000	4,400	5,000	4,400	5,000	4,400
Clusters	3,500	2,800	3,500	2,800	3,500	2,800
Wald statistic					0.83	0.85
C. Rental income						
Incentive	-0.0135	-0.0920	-0.0235	-0.0825	-0.0290	-0.0862
	(0.1593)	(0.1678)	(0.1583)	(0.1665)	(0.1591)	(0.1670)
$sinh^{-1}$ admin rental income	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,200	1,900	2,200	1,900	2,200	1,900
Clusters	1,500	1,200	1,500	1,200	1,500	1,200
Wald statistic					1.26	1.03
D. Retirement income						
Incentive	0.0189	0.0628	0.0303	0.0741	0.0331	0.0776
	(0.0480)	(0.0502)	(0.0474)	(0.0496)	(0.0472)	(0.0494)
$sinh^{-1}$ admin retirement income	Yes	Yes	Yes	Yes	Yes	Yes
Observations	9,500	8,400	9,500	8,400	9,500	8,400
Clusters	5,200	4,300	5,200	4,300	5,200	4,300
Wald statistic					2.59***	2.21***
E. SSA income						
Incentive	0.0254	0.0241	0.0291	0.0275	0.0291	0.0266
	(0.0201)	(0.0214)	(0.0197)	(0.0210)	(0.0197)	(0.0210)
$sinh^{-1}$ admin SSA income	Yes	Yes	Yes	Yes	Yes	Yes
Observations	22,000	19,500	22,000	19,500	22,000	19,500

Clusters	10,000	8,100	10,000	8,100	10,000	8,100
Wald statistic					5.24***	4.25***
Controls						
Wave 2	Yes	Yes	Yes	Yes	Yes	Yes
X_{it}			Yes	Yes	Yes	Yes
$\overline{Z}_{\iota}$					Yes	Yes
Weights						
Cross sectional	Yes		Yes		Yes	
Longitudinal		Yes		Yes		Yes

Source: Author's calculations from the linked Survey of Income and Program Participation 2014, Waves 1 and 2, with 1040 tax returns for interest, dividend, and gross rental income, the 1099-R for retirement income, and with the Master Beneficiary Record for Old-Age, Survivors, and Disability Social Security Income in 2013 and 2014.

Notes: Sample is restricted to individuals aged 15 and older and includes imputed responses. We also drop individuals with entirely imputed records (Type Z). The asset income variable regressions are also conditional on the respondent or the respondent's spouse having a PIK, and the other variables are conditional on the respondent having a PIK. All specifications in Panels A – E condition on both their corresponding survey and administrative income amounts to be positive. In addition to the core set of time average of time-varying covariates that is consistent across all regression tables and includes wave dummy, response indicator, interview mode indicator, log of number of household members aged 15 and older, age, age squared, educational attainment indicators, and marital status indicators, each panel includes the time average of inverse hyperbolic sine transformation of their corresponding administrative income amount. Standard errors are in parentheses and clustered at the household level. Sample size and number of clusters are rounded according to Census disclosure avoidance policies. ****p≤0.01; **p≤0.05; *p≤0.1*.

Table 10. SNAP Measurement Error Regression

	(1)	(2)	(3)	(4)	(5)	(6)
Variables	Measurement Error		Type I Error		Type II Error	
Incentive	0.0045	0.0523	-0.0025	-0.0019	-0.0115	0.0014
	(0.0731)	(0.0786)	(0.0018)	(0.0019)	(0.0169)	(0.0186)
Controls						
include (Wave 2, X_{it}, $\overline{Z}_i$)	Yes	Yes	Yes	Yes	Yes	Yes
$sinh^{-1}$ admin SNAP benefits	Yes	Yes				
if admin SNAP coverage = 0			Yes	Yes		
if admin SNAP coverage = 1					Yes	Yes
Weights						
Cross sectional	Yes		Yes		Yes	
Longitudinal		Yes		Yes		Yes
Observations	2,600	2,300	26,000	22,000	5,400	4,600
Clusters	1,600	1,300	9,500	7,500	2,500	2,000
Wald statistic	2.09**	1.69*	1.67*	1.59*	1.16	1.08

Source: Author's calculations from the linked Survey of Income and Program Participation 2014, Waves 1 and 2, and state-level records on Supplemental Nutrition Assistance Program in 2013 and 2014.

Notes: Sample is restricted to individuals aged 15 and older and includes imputed responses. Sample is also restricted to individuals who reside in a state with SNAP administrative data coverage. We also drop individuals with entirely imputed records (Type Z) and who do not have a PIK. Regressions in Columns (1) and (2) are conditional on positive SNAP benefits in both SIPP and state administrative records. The time-varying observable individual characteristics in these specifications are consistent with the core set of time-varying covariates used in all the other regression tables (wave dummy, proxy response indicator, interview mode indicator, log number of household members aged 15 and older, age, age squared, educational attainment indicators, and marital status indicators), but the specifications in Columns (1) and (2) also include the inverse hyperbolic sine transformation of state-level administrative SNAP benefit amounts. Standard errors are in parentheses and clustered at the household level. Sample size and number of clusters are rounded according to Census disclosure avoidance policies. ***$p \leq 0.01$; **$p \leq 0.05$; *$p \leq 0.1$.

Table 11. Earnings Measurement Error Regression-Arc Change Specification

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Variables								
Incentive	0.0021	0.0023	-0.0001	0.0007	-0.0002	0.0003	-0.0002	0.0004
	(0.0044)	(0.0047)	(0.0042)	(0.0045)	(0.0040)	(0.0043)	(0.0040)	(0.0042)
Controls								
Wave 2	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
$sinh^{-1}$(DER earnings amounts)			Yes	Yes	Yes	Yes	Yes	Yes
X_{it}					Yes	Yes	Yes	Yes
If SIPP earnings ≥ 0	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
$\bar{Z}_\iota$							Yes	Yes
Weights								
Cross sectional	Yes		Yes		Yes		Yes	
Longitudinal		Yes		Yes		Yes		Yes
Observations	49,500	41,500	49,500	41,500	49,500	41,500	49,500	41,500
Clusters	20,000	15,000	20,000	15,000	20,000	15,000	20,000	15,000
Wald statistic							7.97***	7.42***

Source: Authors' calculations from the linked Survey of Income and Program Participation 2014, Waves 1 and 2, with the Detailed Earnings Records in 2013 and 2014.

Notes: Sample is restricted to individuals aged 15 and older. It excludes individuals with entirely imputed records (Type Z), imputed jobs, and those who do not have a PIK. But we include respondents who worked for an employer, were self-employed, or had another work arrangement. All specifications condition on having positive earnings in both SIPP and the DER. In addition to the core set of time-varying observable individual characteristics that are common across every regression table (wave dummy, proxy response indicator, interview mode indicator, log number of household members aged 15 and older, age, age squared, educational attainment indicators, and marital status indicators), all earnings regressions in Columns (5) – (8) including the ones reported in this table include log number of jobs, and indicators for hourly pay, extra earnings, and business earnings. Some specifications in this table also include the inverse hyperbolic sine transformation of earnings in the DER. Standard errors are in parentheses and clustered at the household level. Sample size and number of clusters are rounded according to Census disclosure avoidance policies. ****p≤0.01; **p≤0.05; *p≤0.1*.

Table 12. Asset and Retirement Income Measurement Error Regression-Arc Change Specification

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
A. Interest income								
Incentive	-0.0225*	-0.0222	-0.0231*	-0.0232*	-0.0226*	-0.0231*	-0.0230*	-0.0234*
	(0.0129)	(0.0137)	(0.0128)	(0.0136)	(0.0128)	(0.0136)	(0.0128)	(0.0135)
$sinh^{-1}$admin interest income			Yes	Yes	Yes	Yes	Yes	Yes
Observations	13,000	11,000	13,000	11,000	13,000	11,000	13,000	11,000
Clusters	8,400	6,600	8,400	6,600	8,400	6,600	8,400	6,600
Wald statistic							3.45***	3.53***
B. Dividend income								
Incentive	-0.0113	-0.0177	-0.0130	-0.0181	-0.0138	-0.0190	-0.0136	-0.0187
	(0.0209)	(0.0223)	(0.0208)	(0.0222)	(0.0208)	(0.0221)	(0.0208)	(0.0221)
$sinh^{-1}$admin dividend income			Yes	Yes	Yes	Yes	Yes	Yes
Observations	5,000	4,400	5,000	4,400	5,000	4,400	5,000	4,400
Clusters	3,500	2,800	3,500	2,800	3,500	2,800	3,500	2,800
Wald statistic							0.94	0.98
C. Rental income								
Incentive	-0.0292	-0.0304	-0.0284	-0.0296	-0.0281	-0.0271	-0.0323	-0.0311
	(0.0239)	(0.0250)	(0.0239)	(0.0250)	(0.0238)	(0.0252)	(0.0239)	(0.0252)
$sinh^{-1}$ admin rental income			Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,200	1,900	2,200	1,900	2,200	1,900	2,200	1,900
Clusters	1,500	1,200	1,500	1,200	1,500	1,200	1,500	1,200
Wald statistic							1.36	1.13
D. Retirement income								
Incentive	0.0042	0.0085	0.0057	0.0097	0.0082	0.0128	0.0085	0.0132
	(0.0140)	(0.0146)	(0.0140)	(0.0145)	(0.0139)	(0.0144)	(0.0139)	(0.0144)
$sinh^{-1}$ admin retirement income			Yes	Yes	Yes	Yes	Yes	Yes
Observations	9,500	8,400	9,500	8,400	9,500	8,400	9,500	8,400

Clusters	5,200	4,300	5,200	4,300	5,200	4,300	5,200	4,300
Wald statistic							3.69***	3.68***
E. SSA income								
Incentive	-0.0001	-0.0010	0.0009	0.0013	0.0013	0.0018	0.0014	0.0016
	(0.0052)	(0.0054)	(0.0049)	(0.0051)	(0.0049)	(0.0051)	(0.0048)	(0.0050)
$sinh^{-1}$ admin SSA income			Yes	Yes	Yes	Yes	Yes	Yes
Observations	22000	19500	22000	19500	22000	19500	22000	19500
Clusters	10000	8059	10000	8059	10000	8059	10000	8059
Wald statistic							4.60***	3.55***
Controls								
Wave 2	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
X_{it}					Yes	Yes	Yes	Yes
$\bar{Z}_\iota$							Yes	Yes
Weights								
Cross sectional	Yes		Yes		Yes		Yes	
Longitudinal		Yes		Yes		Yes		Yes

Source: Author's calculations from the linked Survey of Income and Program Participation 2014, Waves 1 and 2, with 1040 tax returns for interest, dividend, and gross rental income, the 1099-R for retirement income, and with the Master Beneficiary Record for Old-Age, Survivors, and Disability Social Security Income in 2013 and 2014.

Notes: Sample is restricted to individuals aged 15 and older and includes imputed responses. Individuals with entirely imputed records (Type Z) are dropped. The asset income variable regressions are also conditional on the respondent or the respondent's spouse having a PIK, and the other variables are conditional on the respondent having a PIK. All specifications in Panels A – E condition on both their corresponding survey and administrative income amounts to be positive. In addition to the core set of time average of time-varying covariates that is consistent across all regression tables and includes wave dummy, response indicator, interview mode indicator, log of number of household members aged 15 and older, age, age squared, educational attainment indicators, and marital status indicators, each panel includes the time average of inverse hyperbolic sine transformation of their corresponding administrative income amount. Standard errors are in parentheses and clustered at the household level. Sample size and number of clusters are rounded according to Census disclosure avoidance policies. ****p≤0.01; **p≤0.05; *p≤0.1.*

Table 13. SNAP Measurement Error Regression-Arc Change Specification

	(1)	(2)	(3)	(4)
Variables				
Incentive	0.0097	0.0128	0.0078	0.0104
	(0.0207)	(0.0225)	(0.0204)	(0.0221)
Controls				
include (Wave 2, X_{it},$\bar{Z}_\iota$)	Yes	Yes	Yes	Yes
$sinh^{-1}$ admin SNAP benefits			Yes	Yes
Weights				
Cross sectional	Yes		Yes	
Longitudinal		Yes		Yes
Observations	2,600	2,300	2,600	2,300
Clusters	1,600	1,300	1,600	1,300
Wald statistic	2.08**	1.95**	2.27***	2.15***

Source: Author's calculations from the linked Survey of Income and Program Participation 2014, Waves 1 and 2, and state-level records on Supplemental Nutrition Assistance Program in 2013 and 2014.

Notes: Sample is restricted to individuals aged 15 and older and includes imputed responses. Sample is also restricted to individuals who reside in a state with SNAP administrative data coverage. We also drop individuals with entirely imputed records (Type Z) and who do not have a PIK. All specifications condition on having positive SNAP benefits in both the SIPP and the state administrative records. Standard errors are in parentheses and clustered at the household level. Sample size and number of clusters are rounded according to Census disclosure avoidance policies. ****p≤0.01; **p≤0.05; *p≤0.1*.